Big Data Cup 2026 (Rough Draft v1)

Forechecking Pressure Topology

Working Draft

February 2026

Question. Can we model forechecking as a continuous pressure field (rather than a fixed system label) and use that geometry to predict whether the opponent exits cleanly, is forced to dump out, or turns the puck over?

Data + scope. 10 games from the public BDC 2026 release, 5v5 only. Rule-based segmentation yields 2,086 forecheck episodes and 14,921 sampled tracking frames (frame stride = 20).

1. Approach Summary

For each episode, we identify the puck carrier each sampled frame and compute an ETA-weighted kernel pressure field from forecheckers:

$$P(x) = \sum_i \exp\left(-\frac{\|x - x_i\|^2}{2\sigma^2}\right) \exp\left(-\frac{\mathrm{ETA}_i(x)}{\tau}\right).$$

We then extract:

- *Local pressure:* mean/peak pressure at carrier, defender proximity, overlap.
- *Topology:* integrated corridor costs (middle, strong boards, weak boards, behind net), most-open corridor cost, and 3s corridor closure rate.
- *Shape and squeeze:* pressure compactness area and funnel-to-boards signal (pressure gradient orientation).

Episode-level signatures are clustered (k-means, $k = 4$) and tested in outcome models (logistic + gradient boosting) against distance/count baselines.

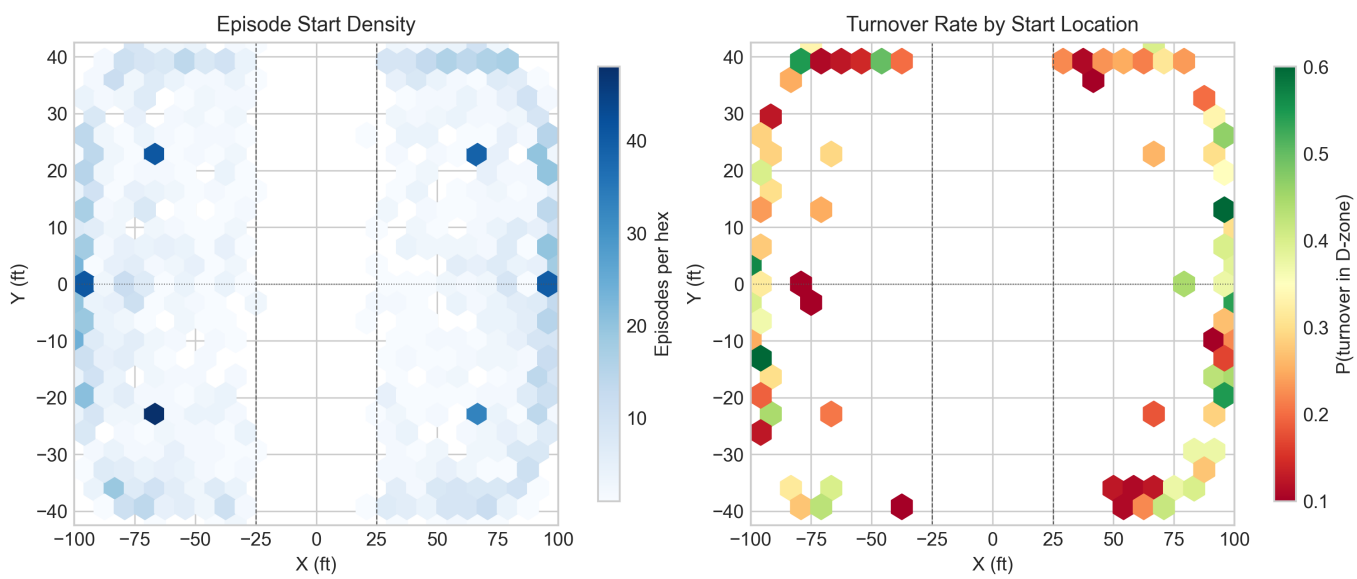

Figure 1: Where forecheck episodes begin, and where they are most likely to become D-zone turnovers. High-risk turnover pockets appear in deep corners/half-wall retrieval zones.

2. Core Findings

Outcome mix. Across episodes: controlled exits allowed 39.9%, forced dump-outs 28.6%, D-zone turnovers 26.1%.

Pressure intensity matters. In the top pressure quartile, denial events (turnover + forced dump-out) occur on 62.4% of episodes versus 46.8% outside that quartile.

Topology improves prediction. Pressure features outperform simple baselines for turnover prediction:

- Gradient boosting with pressure topology: AUC $= 0.758 \pm 0.019$.
- Nearest-defender baseline: AUC $= 0.640 \pm 0.025$.
- Defender-count baseline: AUC $= 0.619 \pm 0.031$.

Archetypes are not classic labels. Cluster C0 is the "tight-net squeeze" archetype: high sustained pressure, high compactness, and the best denial profile (33.2% turnovers, 32.0% forced dump-outs, 29.9% controlled exits allowed).

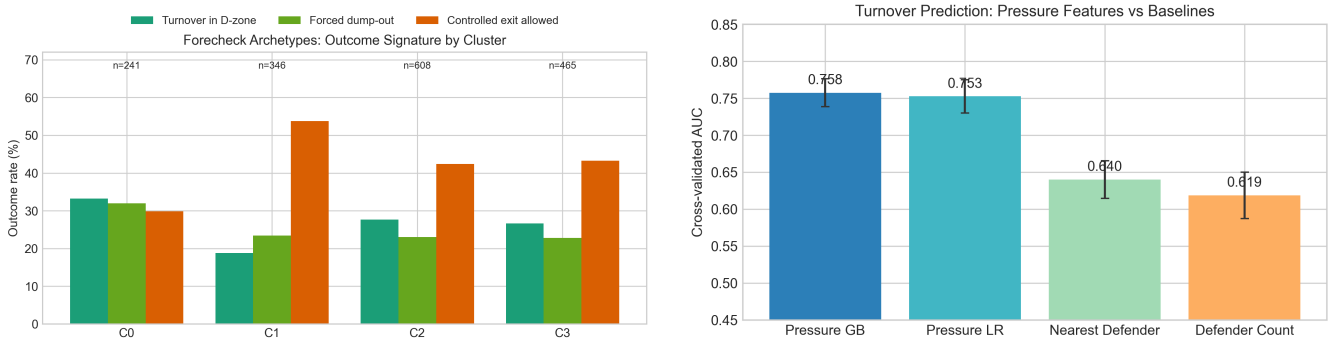

Figure 2: Left: outcome signature by pressure archetype cluster. Right: predictive lift from pressure topology versus simple forecheck proxies.

3. Actionable Recommendations (GM / Head Coach)

A. Build game-plan around pressure *shape*, not just pressure *count*. Drives that simultaneously (i) raise carrier pressure and (ii) collapse open corridors quickly are most disruptive. Practice objective: faster first-3s corridor closure after retrieval.

B. Prioritize retrieval-zone ambushes. Turnover probability is highest when possession begins deep and near boards; use pre-scout triggers for immediate support layers in those sectors.

C. Track team-level TFPI as a process KPI. Define *Team Forecheck Pressure Index*:

$$\text{TFPI} = 100 \cdot (\text{TurnoverRate} + 0.5 \cdot \text{ForcedDumpRate} - \text{ControlledExitRate}) \,.$$

In this sample, Team I (11.6) and Team L (9.8) lead; Team K (-6.3) and Team B (-4.4) trail.

D. Personnel deployment. Deploy forward lines/defense pairs that consistently produce C0/C3-like signatures in one-goal or protect-lead situations.

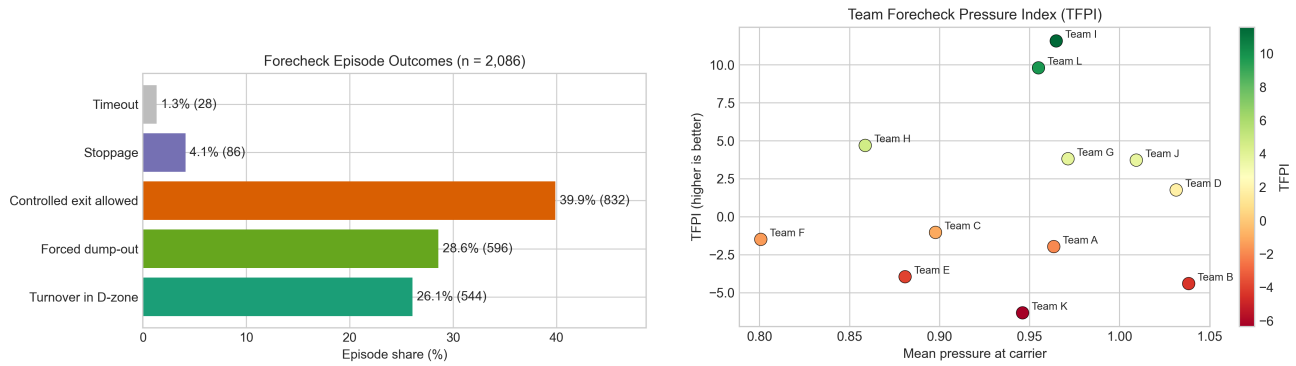

Figure 3: Episode outcome mix (left) and team-level TFPI benchmark (right).

4. Limitations and Next Steps

- Current pressure model is interpretable but not yet acceleration-constrained anisotropic reachability.
- Episode segmentation is rule-based; edge-case validation against video should be added.
- Next revision: infer F1/F2/F3 roles and estimate player-level Pressure Contribution by role.

Code + data appendix path: `projects/forechecking_pressure_topology/`